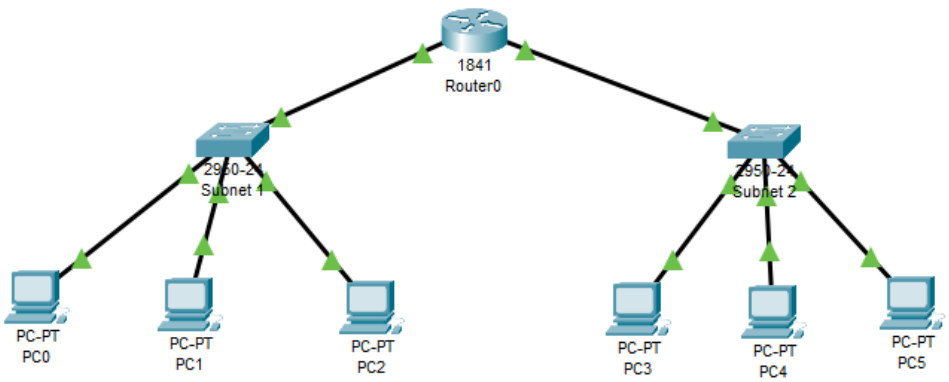
OUTPUT:

1. Subnetting

Subnetting Topology



Ping Verification – Subnetting

- Ping from **PC0** → **PC2** (Same subnet) → Successful

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

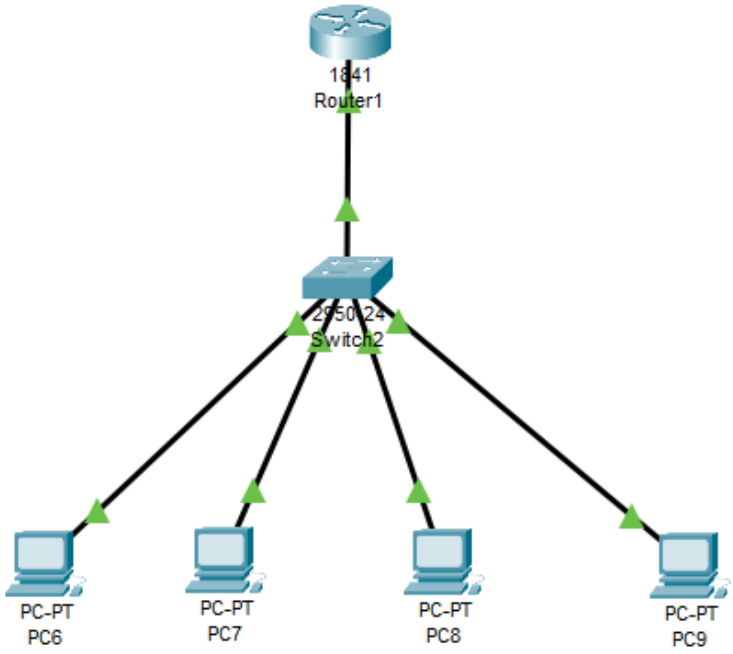
- Ping from **PC0** → **PC5** (Different subnet via router) → Successful

```
C:\>ping 192.168.1.68

Pinging 192.168.1.68 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.68: bytes=32 time<1ms TTL=127
Reply from 192.168.1.68: bytes=32 time=1ms TTL=127
Reply from 192.168.1.68: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.68:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

2. Supernetting

Supernetting Topology

.



Ping Verification – Supernetting

- Ping from **PC6 → PC9** → Successful

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```